Galaxy Life and Death in One Step

Gas supply versus gas depletion timescales set whether galaxies keep forming stars or shut down.

arXiv:2610.20505 — Step-function quenching from gas mass conservation, applied to ~350,000 galaxies.

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Background

Galaxies quench when star formation shuts down, but the many proposed mechanisms—black holes, halos, morphology—have resisted a single unifying description.

No simple framework links diverse quenching processes to one observable quantity across large galaxy samples.

What this paper is selling

Gas mass conservation yields a step-function quenching response governed by supply-versus-depletion competition. The gas-supply timescale, about 1 Gyr, can then be inferred directly from the observed quenched fraction.

Key results to remember

- A characteristic 1 Gyr gas-supply timescale emerges from quenched fractions of ~350,000 galaxies.
- Black holes drive inside-out quenching; dark matter halos drive outside-in quenching.
- Intermediate surface-density galaxies sit at a critical state for quenching and rejuvenation.

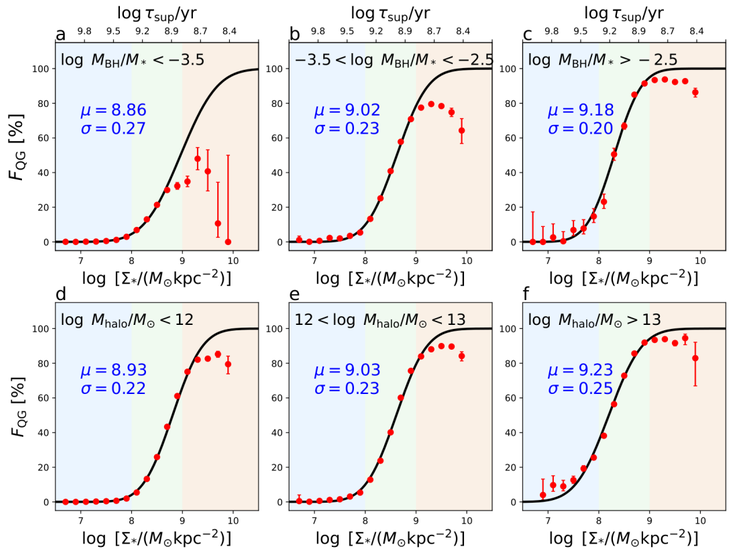


Figure 2 | Black hole growth lengthens and narrows the supply-timescale distribution, while halo mass lengthens but broadens it.

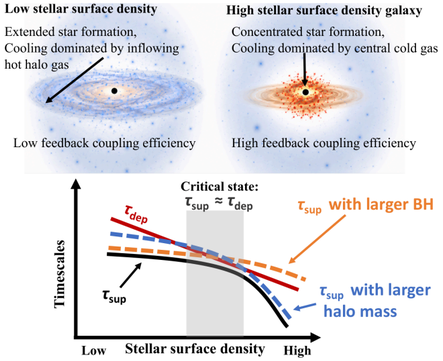


Figure 4 | A physical picture: halo cooling governs extended star formation, while black hole feedback raises supply timescales in compact galaxies.